Term Project Specification Overview

CSSE/MA416 Deep Learning

Goals: For each student
…to work on a project of interest to him/her.
…to gain the opportunity to work on a real-world project on real data.
…to create a hosted web page you can use as part of your portfolio.

Constraints

1. Your "data domain" should be raw structured data like images, sound, language, video. Not data tables.
2. We only have 3 weeks of focused work. So:
    a. Use the types of models similar to those we used in class or that you have some experience with, or plan for extra time to learn how to use them.
    b. Discriminative models (Classification or regression) are easier than generative models.
    c. Generally, transfer learning works best to train the model on new data (instead of using a model without change or training a model from scratch. If you can find a pre-trained model, a far better use of resources that will usually yield higher accuracy is to do transfer learning on an existing model!
3. You'll create a prototype webpage / application that you'll demo to the class. For full credit, you will host so others could use and can be part of your portfolio
4. Past Project Titles for Inspiration.

Milestones and deliverables:

10 points
Week 6: Form project teams. Students will present "lightning talks", a 1-slide presentation to the class, with your name, title, picture, and link to a source of data, submitted in ppt or google-slide-link to the Moodle dropbox. If you have already agreed to work with at least one other person, you only need to present one idea between you. Best case is that you find a whole team before this date and you just present your slide to the class to let them know what you are doing. Otherwise, the purpose of the presentations is to find a complete team (typically 4 people) to work on a mutually-agreeable topic, which you'll do in class on that day. The grade will be 10/10 if your slide contains the required information in the required format and you or an already-agreed-upon team member presents it in class.

Each of the following has a starting template file with further information.

Furthermore, here are the exact grading rubrics used to evaluate each deliverable.

25 points
Week 6: Perform a mini-literature review (details and template, past student example; rubric link above) to see what research has been done on the topic. Your overall goal is to start thinking about your project

in terms of what's been done and what the challenges will be. Specific goals: To read a small number of papers relevant to your term project, understand the approach that others have taken, and start to see how to apply these approaches to your project. If the work cannot be applied to your project in some way, this is an opportunity to see how you'll need to extend someone else's work. Each student should each take the lead in reading and summarizing a paper for the team, but (1) all are responsible for reading / understanding / editing all the summaries, so that (2) the whole team can read more closely whatever paper(s) the team determines to be most relevant, so you all understand the key issues. (You'll find that Not All Papers Are Created Equal. To use an analogy, all teammates should spend some time digging through some "dirt" in search of a gem. But if the team finds a gem, all should study it closely.)

The highest quality places to search are peer-reviewed sources like *journals* (like Pattern Recognition or IEEE Transactions on Pattern Analysis and Machine Intelligence) or c*onference proceedings* (like ICPR = International Conference on Pattern Recognition, ICML = International Conference on Machine Learning, or NIPS – Neural Information Processing Systems). You have access, through Logan Library, to all of these journals and conference proceedings, so don't pay for papers and don't worry if they all aren't generally available online - just login to the Library and check out the ACM Digital Library or IEEE Explore!

30 points
Week 7: Create a Project Baseline (details and template; rubric link above) to plan for the next 3 weeks of work. Main idea: you should download your data and do some standard **preliminary work like gathering statistics and creating a simple baseline classifier/regressor** to test the viability of your approach. The goal is to understand your data to avoid surprises later! Then you should set goals for yourself and a timeline to accomplish them. The template gives the format and some more details.

Workflow note for this and future milestones:
You are welcome to collaborate as a group using any tool that you like (for example, google docs, google drive, google colab, github).
Best practice for sharing code is github. But usually data files are too big to store in github. One option to store your data is on the department servers. Instead, use the department Slurm[1] server. Many students like to train large models on the server.

We'll give every team a project folder. To see it in your home folder so you can run code there, you need to make a *symbolic link* to it. Open a terminal on the server to your **home folder/directory** (that is the default when you launch a terminal. When I print the working directory to verify it, it shows me:

[1] Technically, slurm only handles authentication. It will route you to one of the department Linux servers (gauss, noether, hinton, gebru) which all share /home and /work folders so you can store and access the same data (including large data files) on any of them. Gebru and noether have beefy GPUs and gauss and hinton have decent CPUs. You don't need to remember these names; you just request GPU or CPU resources and it will route it for you transparently.

~$ pwd
/home/boutell

Then make the symlink. For example, if you are team 17, you'd type:
ln -s /work/cssema416/202710/17 .
This creates a link from another folder (**/work/cssema416/202710/17/**) to some other folder (the dot means the current folder, in this case, your home folder).
Now go back to your home folder and you should see folder 17 in your list! (Ideally, you can edit files that you put there. Jupyter occasionally messes up permissions, so if you can't edit them, then just make them in your home directory and then copy them to your work folder to run and share with your team as needed. And if you need help at any time, please reach out to Darryl Mouck (mouck@rose…), the CSSE/MA System Administrator. He's a great resource for you.)

20 points
Week 8 Sprint report (details and template; rubric link above). Each week, each of you will list your collective and individual accomplishments in a document. It is due on Thursday; Friday in class you will meet as a group with your instructor, although this sometimes changes - see the course calendar for dates/times. Your instructor will walk quickly through your report with you to confirm its contents if needed, asking you to demonstrate any other evidence of work completed if it is not clear, before discussing your questions and next steps.

20 points
Week 9 Sprint report and System Demo (details and template; rubric link above). All the requirements of the week 8 report PLUS you will show your instructor a demo of your live website that does prediction on individual data points (see notes below about the demo). You can still work on polishing it and adding features after this date, but you must have something ready to show in this meeting.

300 points
Week 10: In-class presentation and demonstration of your project (rubric link above). These will be scheduled for the last week of the term. (Your presentation day will be determined in class ahead of time.) **Plan for 15-20 minutes, plus questions**.

Basic flow of presentation: **This presentation and demo are a pitch for you to sell your project to an audience. Focus on how well it works or what is needed to make it better.** So capture interest with a powerful introduction, clearly explain what you are trying to solve (stats about your dataset and an example of a single data record), give 3-5 slides of your workflow with some technical detail (at least your model or models), show 1 slide of overall results, show results on individual examples (this is your online demo), state some obstacles you faced and lessons learned from overcoming them, and conclude with how your work builds on others (adapted from your lit review). This will be a minimum of 7 slides (1 per slide for the first 7 lines on the rubric) and likely several more. There is a lot here for the allotted time so you must practice so that you keep things moving and don't lose time in transitions.

But see the rubric for more details!
For the "demo" portion, show your system operating to predict the target **on selected individual examples**, a mix of successes and failures, then explain the results. The main purpose is to pay attention to results at a finer level of detail than just the aggregate score (m.s.e. or accuracy). You'll want to run the predictions in real-time, which works since the estimator is already trained - you certainly don't need to show the training or hyper-parameter tuning phase running. then try to engage your classmates by asking for them to choose sample inputs or guess at predictions before you run the predictor.

If your demo isn't acceptable to your instructor, you will need to fix it and show it to your instructor in their office during exam week.

For this course, you must store your demo in a permanent location (a webpage on a server), so that you can share a URL to your demo as part of a portfolio of work that you do at Rose-Hulman. How you develop it (HuggingFace, gradio, streamlit, ONNX, flask) and where you host it (HuggingFace, gradio, Firebase, github.io, an RHIT server, or anywhere else) is up to you. Opinions vary about ease of use, some like Huggingface, some say streamlit is easy, others say gradio is easier than streamlit. But you aren't the first ones to want to host a model online, and companies are catering to this need, so take advantage of them. If you have no experience with website creation, this is a valid and suggested use of AI to code it. (If you do have experience, I'll still leave the choice to you - you may want to practice your web skills or you may need to just get it done quickly - either way, the website isn't a Deep Learning course outcome so AI usage is acceptable with attribution.)

You will also submit your project experimentation code itself (zipped or links to a shared space or repo) to another dropbox - as you saw on the rubric, submitting code that is easy to understand and run is part of your project grade. (Be sure to include a detailed README, as described in detail in the rubric.)

20 pts (could be counted in part of attendance grade as participation).
Week 10: Peer evaluation of presentations in-class using the Moodle survey. You will evaluate every project except those on your presentation day, so you aren't distracted from your own presentation.

10 points.
Week 10: Team Accountability Survey and Project Reflection. Completion of this survey is required to pass the project.